

Postpartum Hemorrhage 07: Cesarean, Accreta, and Planned Hemorrhage

Companion reading handout for simultaneous listening.

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Cesarean, accreta, and planned hemorrhage.

Until now, most of this course has lived inside the emergency.

A uterus is soft.

Blood is pooling.

The team is called.

The blood bank is notified.

The operating room is prepared.

But some hemorrhage should be treated before it starts.

That is the theme of this episode.

Accreta is where the best PPH treatment happens before the hemorrhage begins.

And cesarean delivery is where small technical choices can become blood loss choices.

Let us start with ordinary cesarean prevention

After delivery of the infant, oxytocin should be started.

Gabbe describes 10 to 80 international units of oxytocin in 1 liter of crystalloid, infused over 4 to 8 hours, as a prevention strategy for uterine atony and PPH after cesarean.

The key practical point is not the highest dose.

The key point is that oxytocin is the routine prophylactic uterotonic.

Misoprostol should not be used in place of oxytocin for routine cesarean PPH prevention.

Tranexamic acid before incision may reduce blood loss and the need for uterotonics, and carbetocin after delivery may reduce the need for uterotonics and uterine

massage, but oxytocin remains the central routine drug in the source material.

Now placental delivery.

At cesarean, spontaneous placental expulsion with gentle cord traction is preferred over manual removal.

Several trials show less blood loss and less endometritis when the placenta is allowed to separate spontaneously with gentle traction and uterine massage.

That is not a romantic preference.

It is a hemorrhage prevention maneuver.

Placental patience prevents bleeding.

Now the uterine incision.

Blunt uterine incision expansion is preferred.

Gabbe describes blunt extension of the uterine incision using cephalad-caudad traction, meaning upward and downward traction, rather than sharp extension or direct transverse pulling.

Why?

Because sharp extension is associated with more blood loss and transfusion, and transverse expansion is associated with more unintended extension and more blood loss.

So the surgical technique memory is:

Blunt expansion prevents extensions.

Placental patience prevents bleeding.

Incision inspection prevents concealed hemorrhage.

After the infant and placenta are delivered, inspect the uterine incision before closure.

Inferior extensions must be visualized and repaired separately before closure.

Lateral and inferior extensions can endanger the ureter. If bleeding extends into the broad ligament, opening the space and identifying the ureter before suture placement may be helpful.

This is not academic anatomy.

A poorly recognized extension can become concealed hemorrhage.

A hurried suture can create ureteral injury.

An uninspected angle can keep bleeding after the abdomen is closed.

Cesarean PPH causes are not mysterious.

They are atony, incision extension, placental bed bleeding, previa, accreta, broad ligament bleeding, retroperitoneal bleeding, and sometimes uterine rupture or injury to adjacent structures.

The first treatment during cesarean is familiar: uterine massage and uterotonics for atony, fluid and blood products, sequential labs, meticulous repair of uterine incision tears, repair of incision angles and extensions, and hemostasis of bleeding vessels.

If that does not stop bleeding, the local protocol moves to uterine artery or internal iliac artery ligation, B Lynch suture with or without balloon integration, and hysterectomy when bleeding is life-threatening.

Now let us move to placenta previa and accreta

Placenta previa is dangerous because the placenta is over or near the cervical os, and the lower uterine segment is a poor contracting segment.

At cesarean for previa, the surgeon should anticipate rapid blood loss.

Blood products should be crossmatched and available.

Before incising the lower uterine segment, assess the vascularity.

A low transverse incision is not absolutely contraindicated, but a vertical uterine incision may be preferable in selected cases, especially with anterior placenta previa or when placenta accreta is suspected.

Ideally, do not cut through the placenta.

If the placenta is disrupted, delivery should be expedited.

After delivery, if the placenta separates, bleeding is controlled by contraction of the myometrium around spiral arteries. But the lower segment contracts poorly, so placental bed bleeding can be significant.

That is why aggressive uterotonic therapy, surgical intervention, tamponade, or local hemostatic measures may be needed.

Now accreta

Placenta accreta spectrum means abnormal adherence or invasion of the placenta into the uterine wall.

Accreta attaches abnormally.

Increta invades into the myometrium.

Percreta extends through the myometrium and may involve surrounding structures such as the bladder.

The two major risk factors are placenta previa and prior cesarean delivery.

The risk becomes dramatic when they combine.

Gabbe gives the following risk pattern for placenta accreta when placenta previa is present.

With no prior cesarean, the risk is about 3 percent.

With one prior cesarean, about 11 percent.

With two prior cesareans, about 40 percent.

With three prior cesareans, about 61 percent.

With four or more prior cesareans, about 67 percent.

Those numbers are worth hearing because they change how you look at an ultrasound report.

Placenta previa over a scar is not just "a placenta location.

It is a possible planned hemorrhage.

Prenatal diagnosis improves outcomes.

It reduces blood loss and transfusion requirements because the team can prepare.

Ultrasound is the preferred diagnostic modality. Findings may include loss of the normal zone between placenta and myometrium, disruption near the bladder interface, exophytic placental masses, and vascular lacunae.

Color Doppler can add information, and magnetic resonance imaging can help when ultrasound is equivocal, the placenta is posterior, or invasion into surrounding tissue must be assessed.

But for this course, the core teaching is not imaging criteria.

The core teaching is planning.

Accreta is a planning diagnosis.

The ideal management is multidisciplinary.

Maternal-fetal medicine.

Experienced pelvic surgeons.

Obstetric anesthesia.

Blood bank and blood conservation support.

Interventional radiology when appropriate.

Neonatology.

Urology when percreta or bladder involvement is possible.

The delivery should be scheduled when the right people and resources are available, in a facility prepared for major obstetric hemorrhage.

Gabbe notes that many authorities favor delivery around 34 to 35 weeks, depending on clinical circumstances.

Adequate access is mandatory.

At least two large-bore intravenous lines.

Often arterial monitoring, central access, rapid infuser, cell salvage, and abundant blood products are considered according to the situation and institutional practice.

The study manual emphasizes that planned cases may require preparation for 10 to 20 packed cell units, 10 to 20 fresh frozen plasma units, and platelets.

The exact number is less important than the mindset:

Do not discover the blood bank after the placenta starts bleeding.

Now the intraoperative rule.

If accreta is suspected, avoid disturbing the placenta.

Incise the uterus away from the placenta.

Deliver the fetus.

Clamp the cord.

Leave the placenta in situ when proceeding with cesarean hysterectomy or another planned strategy.

Disruption of the implantation site can cause rapid blood loss.

This is the contrast that should stay in your memory.

Scene one:

Unrecognized accreta.

The baby is delivered.

Someone pulls on the placenta.

It does not separate.

Someone pulls harder.

Profuse bleeding starts.

The team is now trying to build a hemorrhage plan inside an active hemorrhage.

Scene two:

Suspected accreta.

The diagnosis was considered antenatally.

The case is scheduled with senior surgeons, anesthesia, blood bank, neonatology, and urology if needed.

Blood products are available.

The incision plan avoids the placenta.

The placenta is left undisturbed.

The surgical plan is executed before the placental bed is torn open.

That is the difference between surprise hemorrhage and planned hemorrhage.

Now conservative accreta management.

Residents may hear "conservative" and think "easy fertility preservation.

That is the wrong emotional tone.

Conservative accreta management can include leaving the placenta in situ, delayed removal, wedge resection or oversewing, lower segment tamponade, curettage, embolization, hemostatic sutures, arterial ligation, and other strategies in selected cases.

But the risks are serious: delayed hemorrhage, infection, fistula, subsequent surgery, hysterectomy, uterine necrosis, recurrent accreta, rupture, and even death.

Conservative management is not casual.

It is selected, planned, monitored, and risk-laden.

Methotrexate is not routinely advocated in the study manual synthesis.

Balloon occlusion and embolization can be useful in selected cases, but the literature and local practice must be interpreted carefully. These are adjuncts, not guarantees.

Now connect accreta back to PPH.

In ordinary atony, the uterus fails to clamp down.

In accreta, the placenta is abnormally attached, and attempts to detach it can open a massive bleeding surface.

In previa, the placental bed is in the lower segment, which contracts poorly.

So the best treatment is anticipation:

Recognize the risk.

Diagnose when possible.

Refer when appropriate.

Plan the team.

Plan the blood.

Plan the incision.

Plan what will happen to the placenta.

Plan the hysterectomy or conservative strategy before the hemorrhage forces the decision.

Now the cesarean hemorrhage script.

Oxytocin after delivery.

Do not replace oxytocin with misoprostol for routine prevention.

Allow spontaneous placental separation with gentle cord traction.

Use blunt cephalad-caudad uterine extension.

Inspect the incision and repair extensions.

If previa or accreta risk is present, blood and senior help must be ready before incision.

If accreta is suspected, do not disturb the placenta.

Incise away from the placenta.

Deliver the fetus.

Proceed according to the planned strategy.

Let us close with the sentence that defines this episode:

Accreta is where the best PPH treatment happens before the hemorrhage begins.

In the next and final episode, we will leave the operating room and return to the patient after the emergency: delayed postpartum hemorrhage, anemia, breastfeeding, fatigue, thrombosis prevention, and the final synthesis of how to think through PPH from first recognition to recovery.